

A target contraction generator lies in the source body

MathlibAnnex.PluckerBody.normalizedGenerator_mem_of_sphereIsometry

theorem

Transfers membership through two extensions with the same boundary trace.

Statement

Let $m, N \in \mathbb{N}_0$, put $n = m + 1 > 0$, and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p, q : E \rightarrow \mathbb{R}$ be continuous real seminorms with specified positive constants a_p, b_p, a_q, b_q such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty \quad (x \in E).$$

Both p and q are norms. For each $s \in \{p, q\}$ define the closed unit ball, unit sphere, and real reference volume by

$$K_s = \{x : s(x) \leq 1\}, \quad \Sigma_s = \{x : s(x) = 1\}, \quad v_s = \lambda_n(K_s) \in (0, \infty).$$

The finite measure is read as a real number when used as a scalar.

Let $\mathcal{I}_{n,N}$ be the finite set of increasing n -tuples of distinct rows from $\{1, \dots, N\}$. The minor-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ carries its sup norm. For a linear map $B : E \rightarrow F$, put $B_{ij} = (Be_j)_i$ in the standard coordinate basis. Write $\Pi_n(B)$ for its maximal-minor vector and $\mathcal{V}_p(B), \mathcal{V}_q(B)$ for its two volume-scaled versions. Their coordinates and scaling are

$$\Pi_n(B)_I = \det(B_{i_a,j})_{a,j=1}^n, \quad \mathcal{V}_s(B) = v_s \Pi_n(B) \quad (s \in \{p, q\}) \quad (I = (i_1 < \dots < i_n)).$$

A p -contraction is a linear map $B : E \rightarrow F$ satisfying $\|Bz\|_\infty \leq p(z)$ for every $z \in E$. Write $\mathcal{C}_p^N$ for these maps. Write $\mathcal{G}_p^N$ for the signed generator set and $\mathcal{P}_p^N$ for its real convex hull, so

$$\mathcal{G}_p^N = \{\mathcal{V}_p(B), -\mathcal{V}_p(B) : B \in \mathcal{C}_p^N\}, \quad \mathcal{P}_p^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_p^N).$$

Let $\Delta : \Sigma_p \rightarrow \Sigma_q$ be a bijection preserving chord distances, so

$$q(\Delta(u) - \Delta(v)) = p(u - v) \quad (u, v \in \Sigma_p).$$

Read its radial extension on the reference coordinates:

$$R_\Delta(0) = 0, \quad R_\Delta(x) = p(x)\Delta(x/p(x)) \quad (x \neq 0).$$

For $x \neq 0$, positivity of $p(x)$ makes $x/p(x) \in \Sigma_p$, so this is defined.

Let $A : E \rightarrow F$ be a q -contraction, meaning $\|Az\|_\infty \leq q(z)$ for all $z \in E$. Then its target generator belongs to the source body:

$$\mathcal{V}_q(A) \in \mathcal{P}_p^N.$$

Assumptions

The dimension is $n = m + 1 > 0$. The norms p, q have the stated continuity and positive comparison constants; Δ is a bijective chord-distance isometry. Here A must be a q -contraction. The integration measure remains fixed, and N is any nonnegative integer.

Conclusion

The positively signed target vector $v_q \Pi_n(A)$ belongs to $\mathcal{P}_p^N$, regardless of the orientation sign of the radial extension.

The historical name “normalized generator” refers here to the volume-multiplied vector $v_q \Pi_n(A)$. It is not division by v_q .

Proof route

Extend the boundary contraction, compare its average with a radial average, then use sign symmetry.

▼ Proof steps

1. **Form the boundary map and extend it.** Let $h(u) = A\Delta(u)$ for $u \in \Sigma_p$. Then

$$\|h(u) - h(v)\|_\infty = \|A(\Delta(u) - \Delta(v))\|_\infty \leq q(\Delta(u) - \Delta(v)) = p(u - v).$$

The contraction hypothesis and chord isometry supply exactly the input of the boundary-extension theorem. Choose its single witness f . For any $k : E \rightarrow F$, write $Dk(x)$ for its Fréchet derivative where it exists and for zero otherwise. Define

$$G_{p,k}(x) = v_p \Pi_n(Dk(x)), \quad \mathcal{A}_p(k) = v_p^{-1} \int_{K_p} G_{p,k}(x) d\lambda_n(x).$$

The five outputs are

$$\begin{aligned} \|f(x) - f(y)\|_\infty &\leq p(x - y), & f|_{\Sigma_p} &= h, \\ Df(x) &\in \mathcal{C}_p^N \text{ a.e. on } K_p, & G_{p,f} &\in L^1(K_p), & \mathcal{A}_p(f) &\in \mathcal{P}_p^N. \end{aligned}$$

2. **Compare two extensions only on the boundary.** Put $H = A \circ R_\Delta$. For every $u \in \Sigma_p$,

$$f(u) = h(u) = A\Delta(u) = A(R_\Delta(u)) = H(u).$$

The equality $R_\Delta(u) = \Delta(u)$ holds because $p(u) = 1$. The maps may differ away from the boundary. Let $\|A\|_{\text{op}}$ denote the operator norm for the two coordinate sup norms. Their global reference-norm estimates are

$$\begin{aligned} \|f(x) - f(y)\|_\infty &\leq b_p \|x - y\|_\infty, \\ \|H(x) - H(y)\|_\infty &\leq \|A\|_{\text{op}} \frac{3b_p}{a_q} \|x - y\|_\infty. \end{aligned}$$

For the second, use linearity of A and the radial 3-Lipschitz estimate with the comparison constants; it is not a 1-Lipschitz assertion for the radial extension.

3. Transfer the minor integrals and normalize. For each $I \in \mathcal{I}_{n,N}$, the boundary comparison gives

$$\int_{K_p} \Pi_n(Df(x))_I d\lambda_n(x) = \int_{K_p} \Pi_n(DH(x))_I d\lambda_n(x).$$

Apply the boundary-equality theorem for maximal-minor integrals with the continuous seminorm p , compact ball K_p , positive dimension $n = m + 1$, maps f, H , and the global Lipschitz bounds of Step 2. Its remaining hypothesis is exactly $f(u) = H(u)$ on Σ_p , also established in Step 2. This produces the displayed equality for the same row selection I . Compact-set integrability of derivative generators applied separately to f and H on K_p justifies their vector integrals. For either $k = f$ or $k = H$,

$$\begin{aligned} (\mathcal{A}_p(k))_I &= v_p^{-1} \int_{K_p} v_p \Pi_n(Dk(x))_I d\lambda_n(x) \\ &= \int_{K_p} \Pi_n(Dk(x))_I d\lambda_n(x). \end{aligned}$$

The cancellation uses $v_p > 0$. Substituting the equal minor integrals yields

$$\mathcal{A}_p(f) = \mathcal{A}_p(H).$$

This is the average comparison from boundary traces; it compares the averages, not the functions away from their common boundary.

4. Insert the signed radial average and remove its sign. The common-sign radial-average theorem supplies $\sigma \in \{1, -1\}$ and

$$\sigma \mathcal{V}_q(A) = \mathcal{A}_p(H) = \mathcal{A}_p(f) \in \mathcal{P}_p^N.$$

Read from the known member at the right: Step 1 gives $\mathcal{A}_p(f) \in \mathcal{P}_p^N$; Step 3 identifies this vector with $\mathcal{A}_p(H)$; the radial-average formula identifies it with $\sigma \mathcal{V}_q(A)$. If $\sigma = 1$, this already gives the conclusion. If $\sigma = -1$, body sign symmetry applied to $-\mathcal{V}_q(A)$ gives $\mathcal{V}_q(A) \in \mathcal{P}_p^N$.

Main citations

- Exact declaration and its source —
MathlibAnnex.PluckerBody.normalizedGenerator_mem_of_sphereIsometry
- the boundary-extension theorem —
MathlibAnnex.Plucker.exists_extension_with_derivativeAverage_mem
- the boundary-equality theorem for maximal-minor integrals —
MathlibAnnex.NullLagrangian.integral_maximalMinor_eq_of_pointwise_boundary_eq
- average comparison from boundary traces — MathlibAnnex.PluckerBody.average_eq_of_trace
- The common-sign radial-average theorem —
MathlibAnnex.Plucker.derivativeAverage_comp_linear_radial
- body sign symmetry — MathlibAnnex.PluckerBody.body_neg
- Compact-set integrability of derivative generators —
MathlibAnnex.Plucker.integrableOn_derivativeGenerator_compact
- Signed generator set — MathlibAnnex.PluckerBody.generators

- Convex-body definition — `MathlibAnnex.PluckerBody.body`

Lean source signature (exact)

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/-- A normalized target generator belongs to the source body under a unit-sphere isometry.
-/
theorem normalizedGenerator_mem_of_sphereIsometry {m N : ℕ}
  {MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)}
  (Δ : sphere (0 : Space MX) 1 ≈i sphere (0 : Space MY) 1)
  (A : (Fin (m + 1) → ℝ) →L[ℝ] (Fin N → ℝ)) (hA : MY.IsContraction A) :
  Matrix.ballVolumeScaledMaximalMinors MY A ∈ body MX N

```

The final colon separates the inputs from the membership conclusion. In particular, `hA` is a hypothesis on `A`, not a new map.

In the source	Mathematical meaning
<code>{m N : ℕ}; Fin (m + 1) → ℝ; Fin N → ℝ</code>	$m, N \geq 0, n = m + 1 > 0, E = \mathbb{R}^n, F = \mathbb{R}^N$. Coordinate vectors have the sup norm; braces mark parameters Lean can infer.
<code>MX; MY</code>	The continuous seminorms $p = \text{MX.p}$ and $q = \text{MY.p}$, with their positive lower and upper comparison constants stated in this Card.
<code>Space MX; Space MY</code>	The same vectors of E , now with norms p and q , respectively.
<code>sphere (0 : Space MX) 1; sphere (0 : Space MY) 1</code>	$\Sigma_p = \{x : p(x) = 1\}$ and $\Sigma_q = \{x : q(x) = 1\}$. The arguments are center 0 and radius 1.
<code>Δ : ... ≈_i ...</code>	A bijective isometry $\Delta : \Sigma_p \rightarrow \Sigma_q$; $q(\Delta u - \Delta v) = p(u - v)$ for all $u, v \in \Sigma_p$. The symbol $\simeq_i$ in the code asserts both bijectivity and distance preservation.
<code>A : ... →_L[ℝ] ...</code>	A continuous real-linear map $A : E \rightarrow F$.
<code>hA : MY.IsContraction A</code>	The target-norm contraction bound $\ Az\ _\infty \leq q(z)$ for every $z \in E$; it is the norm stored in <code>MY</code> , not <code>MX</code> .
<code>Matrix.ballVolumeScaledMaximalMinors MY A</code>	The vector $\mathcal{V}_q(A) = v_q \Pi_n(A)$, where $v_q = \lambda_n(\{x : q(x) \leq 1\})$ and $\Pi_n(A)_I = \det((Ae_j)_{i_a})_{a,j=1}^n$.
<code>generators MX N</code>	$\mathcal{G}_p^N = \{\pm v_p \Pi_n(B) : B : E \rightarrow F \text{ linear}, \forall z, \ Bz\ _\infty \leq p(z)\}$, where $v_p = \lambda_n(\{x : p(x) \leq 1\})$.

In the source	Mathematical meaning
body MX N	The source body $\mathcal{P}_p^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_p^N)$, a set of minor-coordinate vectors.
$\dots \in \text{body MX N}$	The conclusion $\mathcal{V}_q(A) \in \mathcal{P}_p^N$: a vector scaled using the target volume belongs to the body formed using the source norm.

Exact definitions used in this reading: Signed generator set; Convex-body definition.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 15–18:

```

noncomputable section
open Set Metric Function MeasureTheory
open scoped NNReal ENNReal BigOperators
namespace MathlibAnnex.PluckerBody

```

▼ Exact content identity

Declaration: MathlibAnnex.PluckerBody.normalizedGenerator_mem_of_sphereIsometry
 Accepted content SHA-256: d0157f88f3d702b5365e3797e3b8a11c8764945986637091085e6fd4585ebec1
 Accepted source guide SHA-256: bb5e9ff6f537f5d87b15ac86e9a8f0a428a9316f1dd69f711ce424e2ef96083b
 Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73